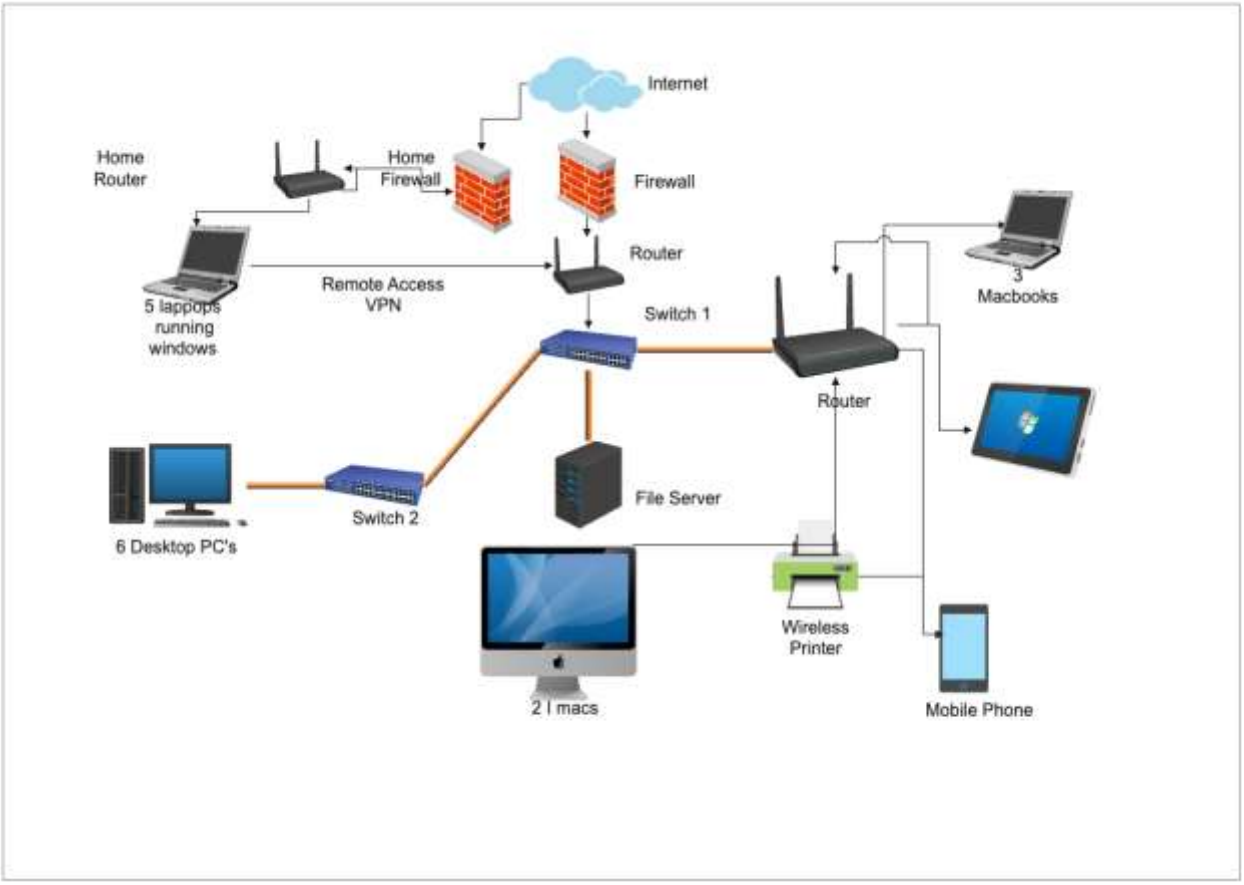
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Task-1 Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

A typical home network can contain a modem, router, switch, wireless access point, and multiple user devices.



Data Flow

Internet → Modem → Router → Switch → Laptop

and

Internet → Modem → Router → Switch → Access Point → Smartphone

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Task-2

List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Device	Main Function	Real-Life Example
Modem	Connects the home network to the Internet Service Provider (ISP) by converting the incoming communication signal into a form the network can use.	Connects the home network to the Internet Service Provider (ISP) by converting the incoming communication signal into a form the network can use.
Router	Connects different networks and forwards data packets to the correct destination. It can also provide NAT, DHCP and often Wi-Fi.	Used at home to share one Internet connection with phones, laptops and smart TVs.
Switch	Connects multiple wired devices within the same local network and forwards Ethernet frames to the appropriate device	Used in an office to connect computers, printers and servers using Ethernet cables.
Access Point (AP)	Provides wireless network access to devices such as smartphones and laptops and connects them to the wired network.	Used in a house, school or office to provide Wi-Fi coverage.
Network Cable	Provides a physical connection for transmitting data between networking devices	Ethernet cable used to connect a computer to a router or switch.

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Task- 3 Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.



1. Ethernet Cable

Description: Ethernet cables such as Cat5e and Cat6 are commonly used to connect computers, routers, switches and other network devices in homes and offices. Ethernet commonly uses an RJ45-style connector.



2. Coaxial Cable

Description: Coaxial cable contains a central conductor surrounded by insulation and shielding. It is commonly used for cable television and cable broadband Internet connections.

Common use: Cable TV, cable Internet and satellite/antenna connections.

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3. Fiber Optic Cable

Description: Fiber optic cable transmits data using pulses of light through glass or plastic optical fibers. It provides high bandwidth and is suitable for long-distance communication.

Common use: Fiber-to-the-home Internet, ISP networks, data centers and Internet backbone connections.

Images and Sources

The following online images can be used as references for the three cable types:

- **Ethernet Cable:** RJ45 Ethernet cable image and information — NETGEAR.
- **Coaxial Cable:** Coaxial cable image and information — Virgin Media.
- **Fiber Optic Cable:** Fiber optic cable image and information — TechTerms.

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Task- 4 Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Difference	Router	Switch
Main Use	Connects different networks and routes packets between them.	Connects devices within the same local network.
Typical Location	Commonly found between the modem/ISP connection and the home or office LAN.	Commonly found inside a LAN, such as an office or server room
Data Handling	Uses IP addresses to determine where packets should go.	Primarily uses MAC addresses to forward Ethernet frames.